Chapter 12, Problem 33P

Problem

A three-phase source delivers 4.8 kVA to a wye-connected load with a phase voltage of 208 V and a power factor of 0.9 lagging. Calculate the source line current and the source line voltage.

Step-by-step solution

Step 1 of 1

$$S = \sqrt{3} V_L I_L \angle\theta$$

$$S = |S| = \sqrt{3} V_L I_L$$

For a Y – connected load,

$$I_L = I_P \ , \quad V_L = \sqrt{3} V_p$$

$$S = 3 V_p I_p$$

$$I_L = I_p = \frac{S}{3V_p} = \frac{4800}{(3)(208)} = 7.69 \ A$$

$$V_L = \sqrt{3} V_p = \sqrt{3} \times 208 = 360.3 \ V \ .$$